Retail Business Performance & Profitability Analysis

Project Phase — Elevate Labs Data Analyst Internship

Introduction

Retail businesses often track revenue closely but have less visibility into which categories are actually profitable and how inventory management affects that profitability. This project analyzes a retail transactional dataset of 73,100 records across five product categories (Groceries, Clothing, Electronics, Furniture, Toys) to uncover profit-draining categories, examine inventory turnover, and identify where procurement decisions are misaligned with actual demand.

Abstract

Using SQL, Python, and an interactive Power BI dashboard, this analysis found that revenue is nearly identical across all five categories (~Rs. 110M each), but estimated profit varies sharply (Rs. 22M–Rs. 44M), showing that revenue alone is a misleading indicator of category performance. A full correlation analysis revealed that Units Ordered has an almost exactly zero correlation with actual demand, and runs consistently 19–20% below Units Sold across every category — indicating a company-wide, demand-blind procurement process rather than an isolated issue. A days-of-supply metric further showed that slower-moving inventory is associated with significantly lower profit capture (Spearman correlation –0.65), a pattern consistent across all five categories.

Tools Used

SQL (via Python/SQLite, run against the full dataset) for aggregation and category-level profitability queries · **Python** (Pandas, Seaborn, Matplotlib, in Google Colab) for correlation analysis and visualization · **Power BI Desktop** for the interactive dashboard, including DAX measures for profit, margin, and inventory turnover.

Steps Involved in Building the Project

1. Data loading and validation — loaded the full 73,100-row dataset into a SQLite database via Python, avoiding the row limits of browser-based SQL sandboxes.

2. Profitability modeling — since the dataset has no cost/COGS column, profit was estimated using documented, category-specific assumed margins (Groceries 20%, Electronics 25%, Furniture 30%, Toys 35%, Clothing 40%), reflecting typical retail margin patterns by category.

3. SQL analysis — calculated revenue and estimated profit by category; investigated per-product inventory patterns (found inventory levels were near-uniformly distributed across products, so the analysis pivoted to category-level and company-wide patterns instead); tested whether Holiday/Promotion and Seasonality affected sales volume (both showed negligible effect); quantified the gap between Units Ordered and Units Sold by category.

4. Python correlation analysis — built a full correlation matrix across all numeric variables, which surfaced the near-zero Units Ordered/demand correlation; calculated a days-of-supply metric and correlated it against estimated profit (Pearson and Spearman), visualized with scatter plots and category-faceted charts.

5. Dashboard design — built an interactive Power BI dashboard (lavender theme) with 4 KPI cards (Revenue, Profit, Margin %, Units Sold), Region/Category/Seasonality slicers, and four charts translating each finding into a clear visual.

Key Findings & Recommendations

Revenue does not equal profit — Furniture had the highest revenue, but Clothing had the highest estimated profit; Groceries produced the lowest profit despite similar revenue to every other category.

Procurement is not demand-driven — Units Ordered showed ~0 correlation with Units Sold and Demand Forecast, and ran 19–20% below Units Sold uniformly across all categories, pointing to a systemic replenishment issue rather than a single category's problem.

Slow-moving inventory loses profit — average profit per transaction fell sharply as days-of-supply increased, consistently across all five categories.

Recommendation — align procurement more closely with Demand Forecast (which correlates almost perfectly with actual sales at 0.997), prioritize shelf space and stock availability for higher-margin categories, and reduce/discount inventory sitting at 10+ days of supply rather than reordering it.

Limitations

Profit figures are estimates based on assumed category margins, not actual cost data — since revenue is nearly equal across categories, the profit ranking largely reflects these margin assumptions and would need real cost data to confirm. The days-of-supply-to-profit relationship is partly mechanical, as Units Sold contributes to both variables; it is best read as a meaningful correlation rather than proof of causation. Per-product inventory levels appeared near-randomly distributed, limiting SKU-level overstock analysis. Finally, the under-ordering finding reflects a gap versus sales, not confirmed lost sales or stockouts, which would require additional data to verify.

Conclusion

This analysis demonstrates that profitability, not raw revenue, should drive category-level business decisions, and that this retailer's procurement process shows a consistent, company-wide disconnect from actual demand. Combining SQL aggregation, Python statistical analysis, and an interactive dashboard produced a coherent, evidence-based narrative connecting profitability, inventory turnover, and procurement behavior — with findings that are both clearly visualized and honestly caveated where the underlying data has real limitations.